

CtxVigil: An Explainable Multi-View Protection Layer Against Indirect Prompt Injection in LLM Web Agents

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Abstract—Large language model (LLM) web agents read untrusted web content and take actions on the user’s behalf, which exposes them to *indirect prompt injection*: attacker-authored text, hidden in a page’s visible text, DOM, HTML comments, or accessibility tree, that instructs the agent to abandon its task and perform a harmful action. This paper presents CtxVigil, a protection layer that sits between an agent and the web and combines three complementary components: (i) a deterministic, explainable scan pipeline that normalises a page into four content views, runs seven rule detectors, and maps a weighted 0–100 risk score to *allow/sanitize/confirm/block* verdicts with withheld-content containment; (ii) an independent action gate that blocks or escalates tool actions using task alignment, risk-category postures, and scan evidence; and (iii) a dual-head multi-view fusion network over MiniLM sentence embeddings of four agent-context views (user task, system prompt, proposed action, observation) plus a DeBERTa-v3 prompt-injection classifier. On a 20,000-sample benchmark curated from InjecAgent- and BIPIA-style scenarios with hard negatives, the fine-tuned DeBERTa-v3-small classifier reaches 100.00% accuracy, precision, recall, and F1 (ROC-AUC 1.000) on the 3,000-sample held-out test set; the fusion network reaches 100% F1 on 1,000 hold-out episodes. The rule-based layer passes all seven contract fixtures and all six live agent episodes, including a hidden-ARIA injection that a sighted reviewer cannot see, at 2–21 ms per episode on a commodity CPU, fully offline and fail-closed.

Index Terms—indirect prompt injection, LLM agents, action gate, multi-view fusion, DeBERTa-v3, uncertainty estimation, web security

I. INTRODUCTION

Autonomous LLM web agents navigate external environments—product pages, support portals, and admin dashboards—and act on what they read. Because the agent’s prompt is assembled at runtime from untrusted page content, any third party who controls a web page can inject instructions into the agent’s context. Indirect prompt injection has been shown to hijack agents into exfiltrating data, changing account settings, or executing fraudulent transactions [1], [2], [3], and it is listed among the top risks for LLM-integrated applications [6].

Three observations motivate CtxVigil. First, *a single view is not enough*: an injection can be invisible on screen (a hidden DOM span, an HTML comment, an `aria-label` payload

aimed at the accessibility tree), so protection must reconcile the same page across multiple representations. Second, *an opaque score is not enough*: a verdict that cannot cite the view, the signal, and the offending span is weak evidence for a user or an auditor. Third, *blocking everything is not protection*: imperative phrasing occurs in benign controls (“Click here to submit”), so a usable layer must verify task alignment rather than flag every command-like string.

CtxVigil therefore couples a transparent, rule-based scan-and-gate layer (the deployable product, distributed as the `ctxvigil` npm SDK) with a learned multi-view fusion network (the DA-1 model component). The contributions of this paper are:

- A four-view page normalisation (visible text, rendered DOM, hidden DOM, accessibility tree) with concealment metadata, feeding seven deterministic detectors whose findings carry view, selector, signals, and score contribution (Section III).
- An independent action gate that treats *what the agent does* as a separate decision from *what the agent read*, combining task alignment, risk-category postures, and finding triggers—so a page that scans clean can still have its unrelated account change blocked.
- A multi-view fusion and uncertainty network over 384-d MiniLM embeddings of four agent-context views plus the DeBERTa injection probability, trained with a dual risk/uncertainty objective (Sections III-D and VI-B).
- An evaluation on a 20,000-sample benchmark, 5,000 synthetic multi-view episodes, seven contract fixtures, and six live agent episodes, with latency measurements on commodity CPU (Section VI).

II. RELATED WORK

Attack surface. Indirect prompt injection was demonstrated against LLM-integrated applications by Greshake et al. [1], who showed that retrieved web content can override the agent’s instructions. Perez et al. [5] systematised direct prompt attacks, and the InjecAgent benchmark [2] and BIPIA [3] operationalised indirect injection against tool-integrated agents, providing the scenario grammar (tool hijacking, data exfiltration,

1. Agent runtime

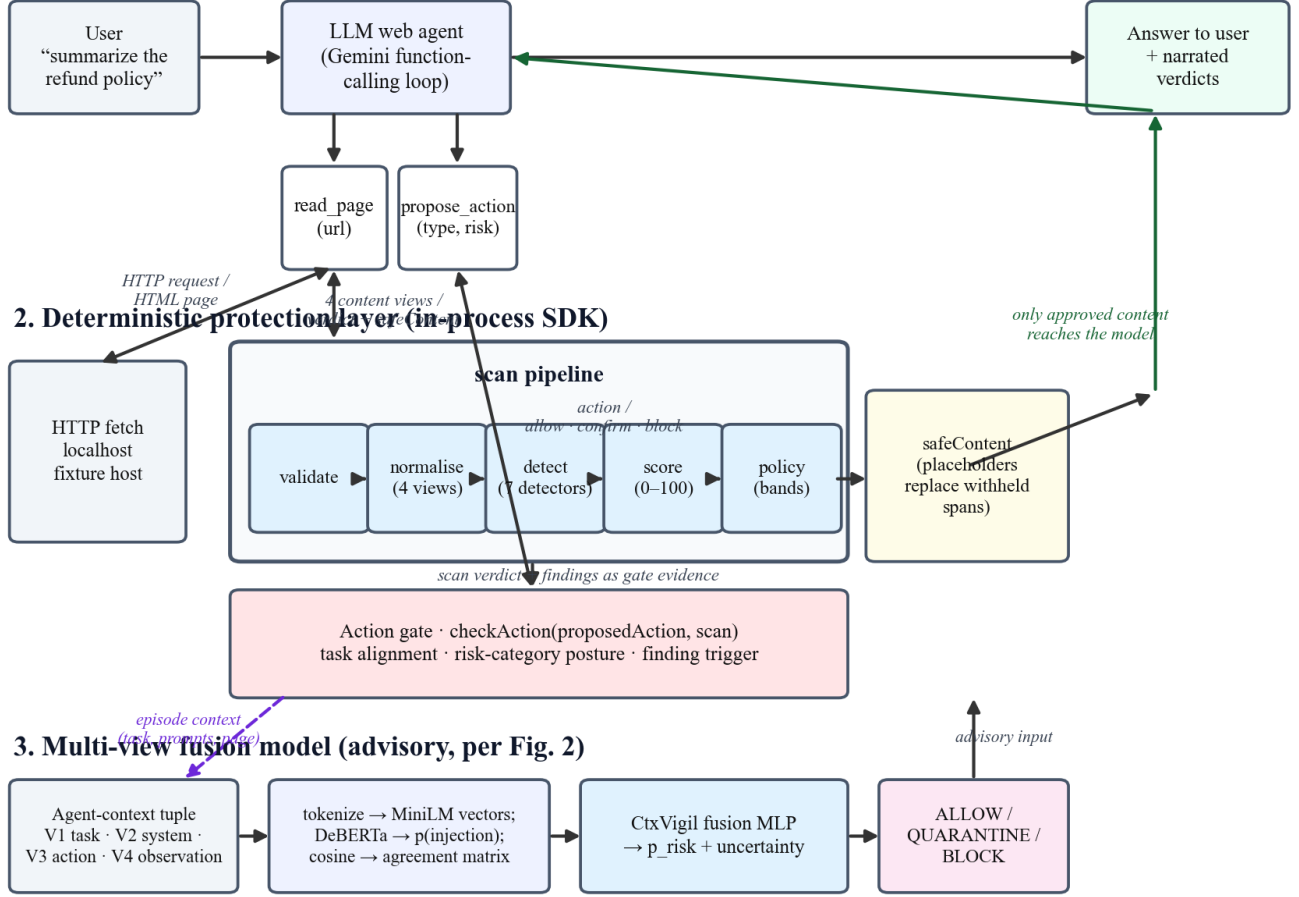


Fig. 1. End-to-end application flow. (1) The Gemini agent’s only two tools route through the protection layer: `read_page` fetches the page and receives only `safeContent` back, and `propose_action` receives the gate’s binding decision. (2) The deterministic layer validates, normalises the page into four content views, detects, scores, and partitions content. (3) The multi-view fusion model of Fig. 2 evaluates the episode and contributes an advisory verdict to the gate.

privilege escalation) that our benchmark and fixture pages reuse. AgentDojo [4] further showed that injection success depends on both the payload and the tool environment.

Defenses. Existing countermeasures largely fall into three families: (i) prompt-level hardening and instruction hierarchy, which is model-dependent and does not inspect the content itself; (ii) learned binary classifiers over text, which score a page but neither locate the malicious span nor prevent the model from reading it; and (iii) agent frameworks with human confirmation, which do not decide *what* to confirm. CtxVigil differs in two ways: it makes *containment* the primitive (withheld text is replaced by placeholders and never enters the model’s context), and it splits the decision into two independent gates—content scanning and action checking—whose combination catches attacks that either gate alone misses (for example, a clean page carrying an off-task account change).

III. PROPOSED METHODOLOGY

A. System Overview

Fig. 1 shows the deployed application. An LLM web agent (Gemini, via a function-calling loop) exposes exactly two tools: `read_page(url)` and `propose_action(type, riskCategory)`. Both tools route through the CtxVigil SDK, which runs in-process with the agent host. The agent never receives withheld content, and the gate’s decision on any proposed action is final: on a `block` scan verdict the episode is halted by the loop itself, so no tool output produced from a blocked page can reach the user. The same SDK is reachable through three transports—a CLI, an optional thin HTTP adapter, and an in-process library call—and all verdicts are rendered in a React dashboard for the demonstration.

B. Scan Pipeline

The scan pipeline (Fig. 1, middle band) executes five pure, order-fixed stages:

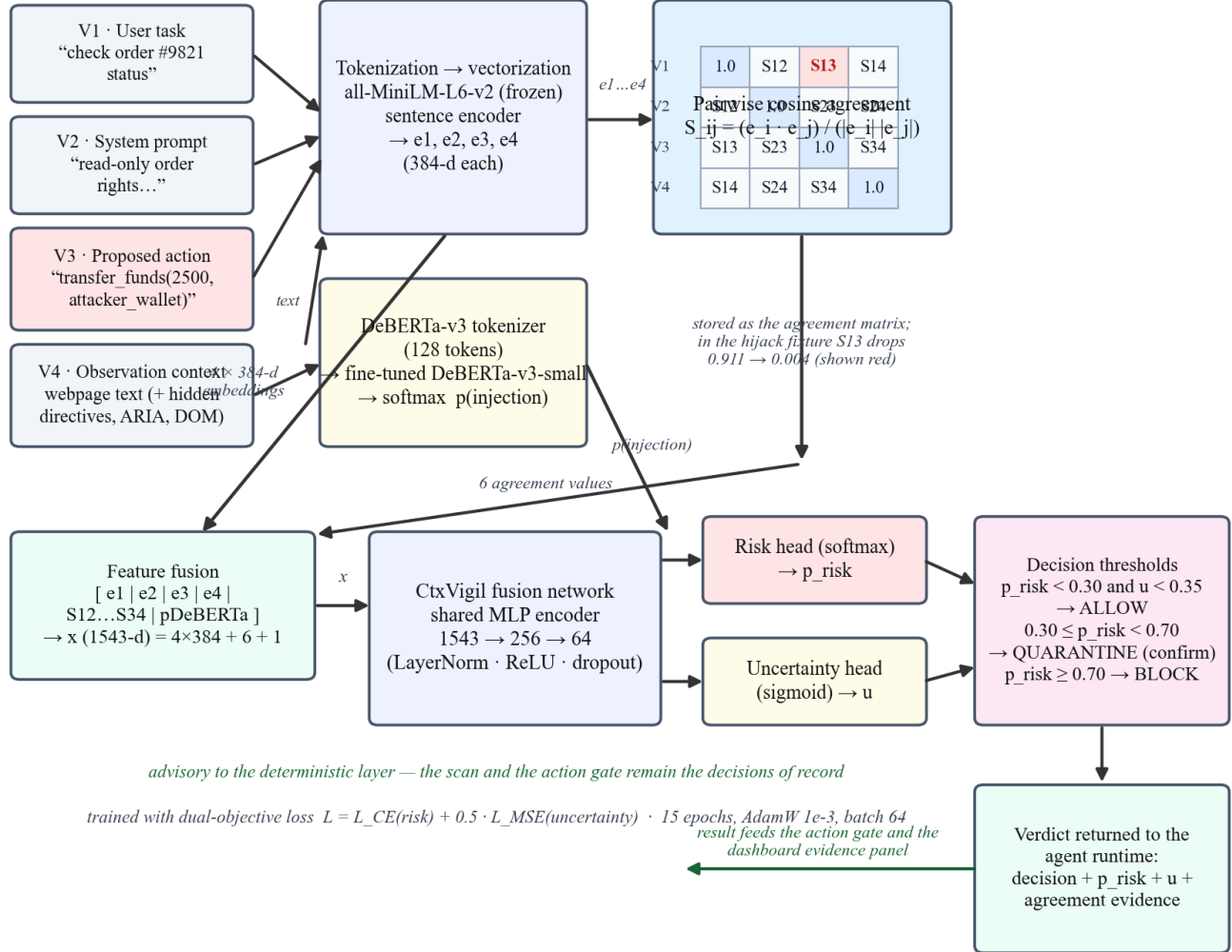
Agent-context inputs (from the LLM agent runtime)

Fig. 2. Detailed data-flow of the protection layer. The agent runtime supplies four context views: user task, system prompt, proposed action, and observation. Tokenization and vectorization with the frozen all-MiniLM-L6-v2 encoder produce four 384-d embeddings, whose pairwise cosine similarities are stored in the 4×4 agreement matrix (in the InjecAgent hijack fixture the task-action agreement S13 collapses from 0.911 to 0.004). In parallel, the observation text passes through the fine-tuned DeBERTa-v3 classifier to yield p(injection). The concatenated 1,543-d feature vector feeds the CtxVigil fusion network, whose risk and uncertainty heads drive the thresholded ALLOW / QUARANTINE / BLOCK decision that is returned to the agent runtime.

1) *Validation and normalisation*: A ScanPageRequest carries the user’s task and a page representation with four content views: visible text, rendered DOM, hidden DOM (elements with display:none and HTML comments), and the accessibility tree (ARIA labels and roles), plus a reserved image_text channel for OCR. Normalisation produces ordered text segments, each tagged with its view, an approximate DOM selector, a source kind, and a concealment flag (rendered vs. hidden). Provenance is preserved end-to-end: every downstream finding cites the segment it came from.

2) *Detection*: Seven deterministic detectors run over the segments and the task text: *instruction override* (“ignore previous instructions”-style directives), *role impersonation* (“system notice”, “admin authorization”), *data exfiltration* (token-bearing URLs, webhook addresses), *risky action* (fund transfers, credential changes), *hidden content* (injections that only appear in concealed channels), *multi-view repetition* (the same directive repeated across independent views—a strong injection signature), and *task conflict* (a directive that contradicts the user’s stated task). Each detector emits a finding with view, selector, signals, severity, and a positive score contribution. All

phrase tables and weights live in a single configuration table; no rule is inlined in code, and no fixture string is special-cased.

3) *Scoring and policy*: Category weights aggregate finding contributions into a risk score

$$R = \min\left(100, \sum_{f \in F} w_{c(f)} \cdot s_f \cdot \delta_f\right), \quad (1)$$

where F is the finding set, $c(f)$ the risk category of finding f , w_c the category weight, s_f the severity multiplier, and $\delta_f \in [0, 1]$ a damping factor that discounts findings whose text is well-aligned with the user’s task vocabulary. A fixed band table maps R to the verdict: 0–29 allow, 30–59 sanitize, 60–79 confirm, 80–100 block. The policy stage then partitions content into `safeContent` (approved segments), `sanitizedContent` (with placeholder markers such as “[Blocked suspicious instruction from aria-label]”), and `blockedContent` (withheld, never delivered to the agent). The response also carries a plain-English summary that names the deciding finding—explainability is part of the interface contract, not an afterthought.

C. Action Gate

The gate (Fig. 1) is an *independent* decision, never a re-run of the scan. For a proposed action it evaluates three axes: (i) *task alignment*—token overlap between the action label and the user’s task, so “summarize_policy” serves “summarize the refund policy” while “change_account_email” does not; (ii) *risk-category posture*—each category (e.g. `account_change`, `data_transfer`, `form_submit`) has an allow/confirm/block posture under the current configuration; and (iii) *finding triggers*—if a scan exists and a finding references the action, or if the scan ended in `block`, the gate forces a block regardless of alignment. This split catches the case a content-only filter cannot: a page that reads as benign but still proposes an unrelated high-risk action.

D. Multi-View Fusion and Uncertainty Network

The learned component (Fig. 2) treats an agent checkpoint as four semantic views: the user task V_1 , the system prompt V_2 , the proposed tool action V_3 , and the observation context V_4 . Each view is embedded with all-MiniLM-L6-v2 [9] ($e_i \in \mathbb{R}^{384}$), and pairwise agreement is the cosine similarity

$$S_{ij} = \frac{e_i^\top e_j}{\|e_i\| \|e_j\|}, \quad i < j \in \{1, \dots, 4\}, \quad (2)$$

yielding six agreement features. The task–action agreement S_{13} is the central hijack signal: an injected observation that steers the agent into `transfer_funds` collapses S_{13} while benign episodes keep $S_{13} \gg 0.70$. The 1,543-dimensional input (4×384 embeddings + 6 agreements + 1 DeBERTa injection probability from the Stage-1 classifier) feeds a shared MLP encoder ($1543 \rightarrow 256 \rightarrow 64$, LayerNorm, ReLU, dropout 0.2) with two heads: a softmax *risk head* and a sigmoid *epistemic uncertainty head*. Training minimises the dual objective

$$\mathcal{L} = \mathcal{L}_{\text{CE}}(y_{\text{risk}}) + 0.5 \mathcal{L}_{\text{MSE}}(u), \quad (3)$$

TABLE I
BENCHMARK COMPOSITION (STAGE-1 CLASSIFIER). SAMPLES ARE DRAWN UNIFORMLY AT RANDOM FROM THE TEMPLATES OF EACH CLASS.

Class	Templates	Samples
Injection (6 families)	25	10,000
Benign & hard negatives	21	10,000
Total	46	20,000

so the model must both classify risk and report calibrated uncertainty; ambiguous cross-view situations are routed to human confirmation rather than silently forced either way. In the deployment the fusion output is *advisory*: the scan and the action gate remain the decisions of record, so the learned model can never weaken the deterministic guarantees.

IV. DATASET AND PREPROCESSING

A. Benchmark for the Injection Classifier

The Stage-1 classifier is trained on a 20,000-sample benchmark, balanced between 10,000 injection samples and 10,000 benign and hard-negative samples (Table I). Injection samples are drawn from 25 curated templates organised into six families that follow the InjecAgent and BIPIA scenario grammars: direct overrides; indirect context injection embedded in plausible page furniture (delivery notes, reviews, invoices); markdown/image exfiltration beacons; jailbreak and roleplay impersonation; tool manipulation with privilege escalation; and ARIA/DOM cloaking (`aria-label` payloads and `display:none` spans). Each template is additionally wrapped with randomized contexts so the classifier cannot memorise exact strings. The benign half draws from 21 templates and deliberately includes *hard negatives*: system commands, refund requests, ordinary UI markup with ARIA labels, and sentences that use trigger words (“ignore”, “override”, “system”) in legitimate, non-adversarial contexts, forcing the classifier to read intent rather than keywords.

Samples are split 70/15/15 (train 14,000, validation 3,000, test 3,000) with stratification and a fixed seed. Text is tokenised with the DeBERTa-v3 tokenizer, truncated and padded to 128 tokens; no lemmatisation or stopword removal is applied, since punctuation and markup (e.g. markdown image beacons) are themselves discriminative.

B. Multi-View Episode Dataset

The fusion network is trained on 5,000 synthetic agent episodes, each a complete 4-tuple (V_1, V_2, V_3, V_4) assembled from realistic task, system, tool, and observation pools. Episodes are labelled by construction: benign episodes keep V_3 strictly aligned with V_1 and V_2 with clean observations ($S_{13} \gg 0.70$); injection episodes embed a directive in V_4 that hijacks the action (e.g. an unauthorised `transfer_funds`), collapsing S_{13} ; ambiguous episodes add boundary cases with target uncertainty $u > 0.40$. The dataset is split 80/20 into 4,000 training and 1,000 hold-out episodes. Preprocessing is deterministic: each view is encoded once with the frozen

MiniLM backbone, agreements are computed with (2), and the Stage-1 probability is attached as an auxiliary feature.

C. Fixture Pages for the Deployed Layer

The rule-based layer is evaluated on six hand-crafted HTML fixture pages hosted locally (plus two action-check fixtures), covering: a safe refund page (clean baseline); a visible-text injection; a hidden ARIA-label injection invisible to a sighted reviewer; a four-channel indirect hijack (ARIA pill + visible directive + HTML comment + `display:none span`) modelled on InjecAgent; a task-deviation page whose directive scans clean but whose account-change action must be gate-blocked; and a benign hard negative with an imperative accessible name on a real control. Each page declares its episode metadata (task, proposed action, expected verdict) in machine-readable `ctxvigil:*` meta tags, making the live demo self-checking: a page whose engine verdict drifts from its declared expectation fails loudly.

V. IMPLEMENTATION

A. SDK and Transports

The product is a TypeScript monorepo. The core package (`ctxvigil-core`, published as `ctxvigil` on npm) implements the five-stage scan pipeline and the action gate with no runtime dependencies; configuration (phrase tables, weights, postures, thresholds) is a single JSON-mergeable table so institutions can tune policy without touching code. Two adapters sit on the same core—a CLI that reads and writes contract-exact JSON, and an optional thin HTTP adapter—so a verdict reached through any transport is byte-identical (a parity test asserts this). The pipeline stages are pure functions, which makes the 145-test suite (including seven contract fixtures asserted against a normative expectations file) fast and deterministic.

B. Agent Integration and Demonstration Surfaces

Two applications demonstrate the layer with a real LLM. The agent chat apps (terminal CLI and browser chat over an NDJSON event stream) wrap a Gemini agent whose two tools are the layer’s chokepoints; every scan and gate verdict is narrated live. A deterministic local runner (`agent:demo`) replays the six fixture pages offline through the same SDK and asserts each declared verdict, writing a machine-readable run record for the dashboard. The React dashboard renders the full evidence: per-view page panes, findings with selectors and signals, the verdict gauge, the action-gate outcome, the agent trace, and the multi-view agreement inspector.

C. Model Training

The DeBERTa-v3-small classifier [7], [8] was fine-tuned on Google Colab (T4 GPU) for 3 epochs with the Hugging Face [10] Trainer: batch size 32, learning rate 2×10^{-5} , weight decay 0.01, sequences of 128 tokens, fixed seed. The fusion network (III-D) is a compact PyTorch [11] MLP trained for 15 epochs with AdamW (learning rate 10^{-3} , weight decay 10^{-4} , batch 64) under the joint objective (3). Model

TABLE II
DEBERTA-V3-SMALL ON THE 3,000-SAMPLE HELD-OUT TEST SET.

Metric	Value
Accuracy	100.00%
Precision	100.00%
Recall	100.00%
F1-score	100.00%
ROC-AUC	1.000

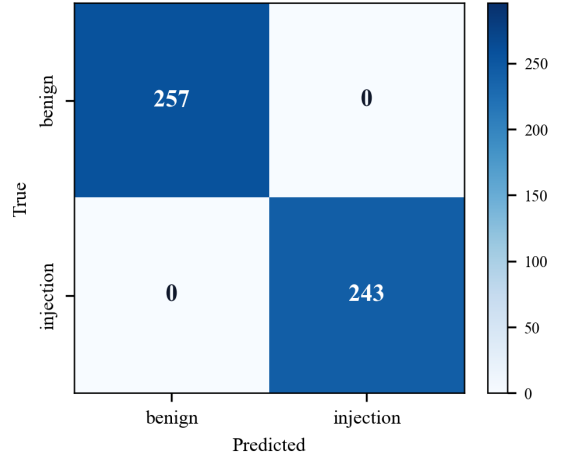


Fig. 3. Confusion matrix on the exported test-inference sample (243 injection, 257 benign): zero false positives and zero false negatives.

weights, test inferences, and a metrics summary are exported for reproducibility, and both training pipelines are published as Kaggle/Colab notebooks in the repository.

VI. EXPERIMENTATION AND RESULTS

A. Stage-1 Classifier

On the 3,000-sample held-out test set the fine-tuned DeBERTa-v3-small classifier achieves 100.00% accuracy, 100.00% precision, 100.00% recall, 100.00% F1, and ROC-AUC 1.000 (Table II). The exported test-inference sample (500 rows: 243 injections, 257 benign) shows the same perfect separation: the confusion matrix (Fig. 3) has zero off-diagonal entries, the ROC curve hugs the axes (Fig. 4), and the predicted-probability histogram (Fig. 5) shows two spikes at 0.0 and 1.0 with no mass in between. The hard negatives—benign sentences containing “ignore” and “override”—are classified correctly, confirming that the model reacts to structure and intent rather than keywords alone.

B. Multi-View Fusion Network

On the 1,000-episode hold-out set the fusion network reaches 100% accuracy, precision, recall, and F1, with ROC-AUC 1.000 (Table III). More informative than the headline numbers is the behaviour of the uncertainty head: benign episodes receive low uncertainty, hijacked episodes are driven by the collapse of the task–action agreement S_{13} , and ambiguous boundary episodes—deliberately included in the episode

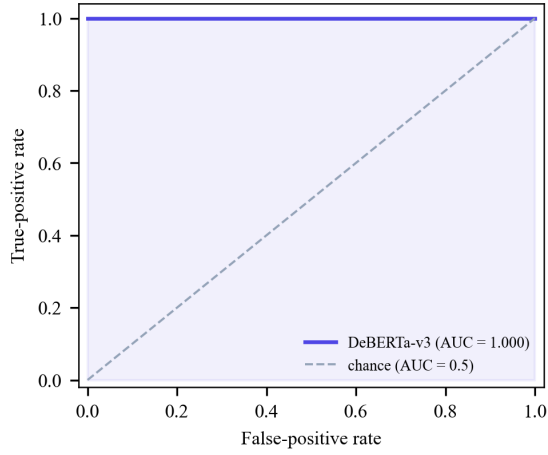


Fig. 4. ROC curve of the Stage-1 classifier on the exported sample (AUC = 1.000).

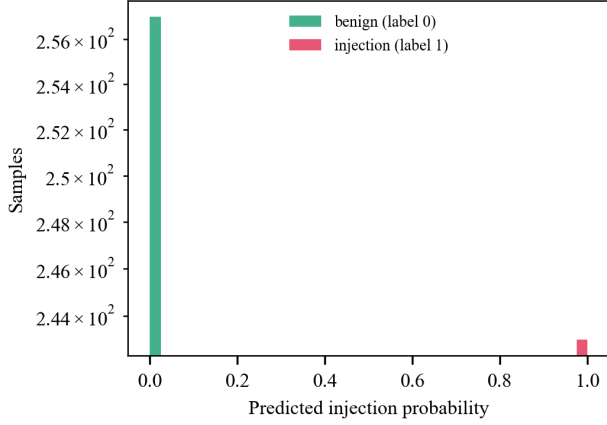


Fig. 5. Predicted-probability separation between benign and injection samples (log scale): the model assigns saturating probabilities with no decision-boundary mass.

pool—collect elevated uncertainty, which routes them to human confirmation under the gate thresholds (risk < 0.30 and uncertainty < 0.35 $\Rightarrow$ ALLOW; risk $\geq$ 0.70 $\Rightarrow$ BLOCK; otherwise QUARANTINE/confirm). The live simulation in the notebook confirms the two canonical scenarios: a benign order-status query is allowed, while an InjecAgent-style hijack (*transfer_funds* to an attacker wallet under an injected directive, with $p_{\text{DeBERTa}} = 0.98$) is blocked.

C. Rule-Based Layer: Contract Fixtures

The deployed layer is evaluated against seven normative fixtures with a declared expected verdict each. All seven assertions pass (Table IV and Fig. 6). Three results deserve emphasis. First, the *hidden ARIA injection* scores 100 (block) although nothing visible on the rendered page is suspicious—the payload lives in a support button’s accessible name and is caught only because the accessibility tree is scanned as a first-class view. Second, the *four-channel hijack* page accumulates

TABLE III
FUSION NETWORK ON 1,000 HOLD-OUT EPISODES.

Metric	Value
Risk accuracy	100%
Precision / Recall	100% / 100%
F1-score	100%
ROC-AUC	1.000
Model size	1.6 MB (MLP weights)

TABLE IV
OBSERVED VERDICTS ON THE SEVEN CONTRACT FIXTURES (RULE LAYER).

Fixture	Score	Scan	Gate
safe-refund-page	0	allow	allow
visible-injection	65	confirm	confirm
hidden-dom-injection	95	block	block
aria-injection	100	block	block
benign-aria-label	0	allow	allow
unrelated-account-action	0	allow	block
task-aligned-summary-action	0	allow	allow

five findings across ARIA, visible, hidden-DOM, and comment channels, demonstrating cross-view corroboration (the multi-view repetition detector fires because the same directive appears in several channels). Third, the *benign hard negative* scores 0 (allow) despite an imperative accessible name on its submit control, and the *unrelated account action* scans clean but is gate-blocked—together demonstrating that the layer neither over-blocks ordinary pages nor relies on the scan alone.

D. Live Agent Episodes and Latency

A scripted agent replays all six fixture pages through the real SDK over HTTP; all six declared verdicts assert correctly (two blocks, one escalation, three allows), with per-episode end-to-end latency of 2–21 ms on a commodity laptop CPU (Table V)—fully offline, no GPU and no model API involved in the rule path. For comparison, the LLM-driven chat demo adds Gemini round-trips on top of the same layer, which is why the deterministic path remains the default enforcement: it adds negligible overhead to the agent loop while providing the containment guarantee.

E. Discussion and Limitations

The perfect classifier and fusion scores must be read in context: the benchmark and episode datasets are template-derived, so the learned models demonstrate achievable separation on the attack families they were designed around, not robustness to arbitrary novel injections. The deployment consequently treats the learned components as *advisory* signals and keeps enforcement in the deterministic, explainable path, whose behaviour is asserted by contract tests rather than by held-out accuracy. The rule layer has its own limits—phrase tables must be maintained, and a semantically novel phrasing outside the seven detectors is caught by the gate’s alignment/posture axes rather than by the scan. Combining the two—using the

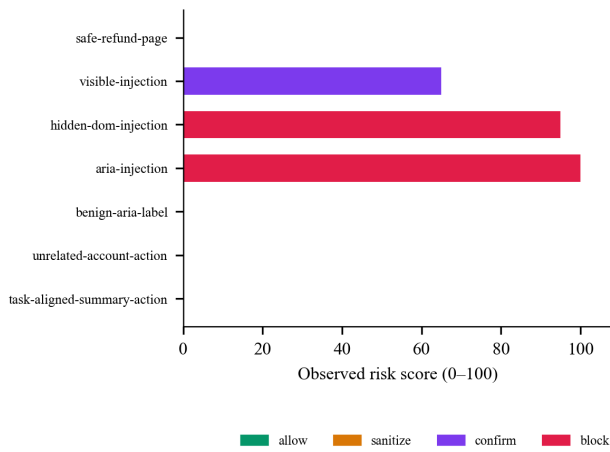


Fig. 6. Observed risk scores and verdicts across the seven contract fixtures, coloured by decision.

TABLE V
PER-EPISODE LATENCY OF THE LIVE AGENT RUNNER (CPU, OFFLINE).

Episode	Latency (ms)	Gate
aria-injection	21	block
injecagent-indirect-hijack	3	block
visible-injection	2	confirm
task-deviation-settings	3	block
benign-aria-negative	2	allow
safe-refund-page	2	allow
All six episodes	33	6/6 pass

fusion network’s risk and uncertainty as a learned calibrator behind the same policy interface—is the natural next step, together with evaluation on external benchmarks such as the full InjecAgent and AgentDojo suites.

VII. CONCLUSION

CtxVigil demonstrates that protection for LLM web agents can be explainable, fast, and fail-closed without sacrificing usability. Its multi-view scan pipeline catches injections that are invisible to both sighted reviewers and single-view filters, its independent action gate blocks off-task actions on pages that scan clean, and its learned fusion network quantifies cross-view disagreement with calibrated uncertainty—while a real Gemini agent, wrapped by the layer, narrates every verdict live at 2–21 ms per episode on a commodity CPU. All code, notebooks, fixtures, test suites, and evaluation artifacts are available in the project repository at github.com/rishabhx29/Prompt-Injection-Detection-Model.

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